UNIVERSITI BRUNEI DARUSSALAM

Triage IQ : Predicting Hypertension from Emergency Department Triage Data

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Declaration of originality

I hereby declare that to the best of my knowledge, this dissertation/report contains no copy or paraphrase of work published by another person, except where duly acknowledged in the text. This dissertation/report contains no material which has been presented for a degree in any other university.

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Abstract

Hypertension emerged as the most frequently recorded comorbidity in Emergency Department (ED) triage records at RIPAS Hospital, Brunei Darussalam, yet no systematic screening for the condition was embedded in the existing triage workflow. This gap was addressed by developing and evaluating machine learning models capable of predicting hypertension risk from structured ED triage data, with the practical aim of making early identification faster and more consistent.

A two-phase approach was used, mainly to separate model selection from real-world validation. In the first phase, seven baseline classifiers which includes Logistic Regression, Random Forest, SVM, and ensemble methods were benchmarked across four publicly available Kaggle datasets related to cardiovascular health and hypertension. XGBoost and Gradient Boosting emerged as the most reliable performers and were selected for the second phase.

The RIPAS dataset brought with it two significant real-world constraints: a severe class imbalance of approximately 96% non-hypertensive records, and a modest total sample of 1,636 patients. SMOTE was applied exclusively to training data to address the imbalance, and both RandomizedSearchCV and GridSearchCV were used for hyperparameter tuning, with recall set as the primary optimization target given the clinical cost of missed diagnoses.

The tuned XGBoost model returned the strongest results with recall of 0.89 and ROC-AUC of 0.945 following threshold adjustment to 0.35. SHAP analysis traced the most influential predictors in the RIPAS context to age, diabetes mellitus, and chronic kidney disease. Overall, the results suggests that machine learning can meaningfully support hypertension screening at the point of ED triage, provided that the data constraints specific to single-center settings are carefully managed.

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Chapter 1: Introduction and Background

1.1 Motivation and Problem

Hypertension is a chronic condition closely linked to elevated risks of cardiovascular disease, stroke, and renal failure. Its clinical significance is well established, what is less well managed is its early detection. In many patients, hypertension produces no symptoms at all in its early stages, which means it can go undiagnosed even during medical contact. The Emergency Department is one such contact point where this gap is particularly visible.

A descriptive analysis of ED triage records from RIPAS Hospital, Brunei Darussalam, made this problem concrete: hypertension was the most frequently recorded comorbidity in the dataset, appearing at a higher rate than any other condition. That finding was the starting point for this study. A condition so prevalent in the triage population was not being specifically screened for during triage, a missed opportunity that an automated predictive model could help address.

Existing triage workflows rely on manual assessment of vital signs, patient history, and presenting complaint. These processes are time-pressured by design and were not built with condition-specific screening in mind. A model trained on triage data already collected during routine care could flag hypertensive patients without requiring any additional data collection or disruption to existing clinical routines.

1.2 Impact, Significance and Contribution

The main result of this study is a validated machine learning framework for predicting the risk of hypertension, based on actual data from the RIPAS Emergency Department. The local data foundation matters: a model trained and tested on Bruneian clinical data is more directly relevant to RIPAS workflows than one generalized from international datasets. Earlier detection of

hypertension at triage could shorten the time between presentation and treatment, reducing the downstream burden of unmanaged hypertension.

The model was design with interpretability in mind from the beginning, rather than incorporating it as an afterthought. SHAP-based analysis was used throughout to ensure that predictions could be traced to specific clinical features, a practical necessity if clinicians are to trust and act on model outputs in time-sensitive triage situations.

1.3 Research Questions

- RQ1: Can machine learning models trained on ED triage data effectively predict hypertension risk among patients presenting to the Emergency Department?
- RQ2: Which features (demographics, vitals, comorbidities) are most important in driving the prediction of hypertension risk in an ED setting?

1.4 Research Objectives

- To benchmark multiple machine learning models on publicly available hypertension-related datasets as a preliminary model selection phase.
- To evaluate the performance of optimized models on the RIPAS ED triage dataset, accounting for real-world data constraints including class imbalance and limited sample size.
- To interpret the best-performing model using SHAP analysis and identify the key predictive features for hypertension risk.

1.5 Scope and Limitations

This study covers both a preliminary benchmarking phase using Kaggle datasets and an optimization phase applied to the RIPAS dataset. Structured triage data was used throughout;

unstructured sources such as clinical notes or imaging were outside the scope of this work. The models were trained and validated on retrospective data from a single hospital, which may constrain generalizability to other clinical settings. Further external validation would be needed before deployment could be considered.

Chapter 2: Literature Review

Hypertension is a medical condition that often goes undetected for years, mainly due to its lack of clear symptoms in the initial stages. This makes it a difficult target for traditional reactive clinical workflows, and has prompted growing interest in whether data-driven approaches, particularly machine learning, could support earlier identification. Emergency Department records and electronic health data have been suggested as promising sources for this kind of screening, since they capture patients at a real point of clinical contact rather than relying on them to proactively seek care.

2.1 Machine Learning Approaches

Logistic regression has long been the standard statistical tool for binary clinical prediction tasks, including hypertension, largely because its outputs are easy to interpret and explain to clinicians. The limitation, however, is that it assumes a linear relationship between features and the outcome, an assumption that does not always hold in real patient data, where variables like age, blood pressure, and comorbidities interact in complex ways. Tree-based ensemble methods such as Random Forests and Gradient Boosting do not carry this assumption, and have shown stronger predictive performance in settings where those interactions matter. Breiman (2001) established the theoretical grounding for Random Forests as an ensemble approach suited to heterogeneous datasets, and the clinical ML literature has broadly supported that finding in practice.

2.2 Use of ED and Triage Data

Triage data has some distinct properties that make it both useful and challenging for predictive modelling. It is collected quickly and under pressure, which means it reflects the acute state of a patient at a specific moment rather than a longitudinal picture. Several studies have shown that

combining ED visit history with current admission data can improve cardiovascular risk prediction accuracy. Choi et al. (2017) demonstrated that recurrent neural networks applied to time-series EHR records could identify early signs of heart failure, which suggests similar temporal approaches might work for hypertension. That said, deep learning methods of that kind typically require large volumes of data to perform reliably, and that is a practical constraint that was directly relevant to this study given the size of the RIPAS dataset.

2.3 Feature Importance and Interpretability

Getting a good prediction is only part of the challenge in clinical machine learning, clinicians also need to understand why a model flagged a particular patient as high risk before they can act on it. Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), which draws on game theory to assign each feature a contribution value for individual predictions. Since its introduction, SHAP has become widely used in clinical ML work because it provides that explanation layer in a way that is both mathematically grounded and relatively communicable to non-technical users.

2.4 Challenges and Gaps

Numerous common challenges can be found in the clinical machine learning literature such as: inconsistent data formats across sites, missing values, class imbalance, and tendency to overfit when working with small datasets. Beyond these general issues, there is a more specific gap relevant to this study, most existing hypertension prediction work has focused on outpatient or general population cohorts rather than Emergency Department data specifically. The ED context is distinct in ways that matter: the data is structured differently, collected under different conditions, and used by clinicians making time-pressured decisions. Studies that have applied ML in ED settings have largely targeted broader outcomes like ICU admission or general deterioration

rather than condition-specific screening (Raita et al., 2019). This study was directly motivated by that gap.

2.5 State of the Art

Risk stratification in ED settings has traditionally relied on scoring tools like NEWS2 (Smith et al., 2013), which is still commonly employed to pinpoint patients who may be at risk of overall deterioration. NEWS2 was not designed with hypertension in mind, however, and shows limited sensitivity for identifying hypertensive patients specifically. Machine learning approaches have been shown to outperform these conventional tools in certain contexts. Raita et al. (2019) found that gradient boosting and XGBoost models outperformed standard five-level triage scoring for predicting critical care outcomes, though that work was focused on general deterioration rather than any specific condition. Chowdhury et al. (2022) reviewed a broader body of hypertension prediction studies and found strong performance from both Random Forest and Logistic Regression models, with age and blood pressure consistently appearing as the strongest predictors, though the reviewed studies drew almost entirely from outpatient or general population datasets, with little ED-specific data represented.

***Table 2.5.1**: Summary of Key Studies in ED Risk Prediction*

Study	Method	Dataset	Key Findings	Problem/gaps
Smith et al. (2013)	NEWS2 scoring system	UK ED cohort	Effective for general deterioration but not hypertension-specific.	Not Hypertension specific
Raita et al. (2013)	Gradient Boosting, XGBoost	Multicenter ED triage dataset	ML models outperformed conventional five-level triage scoring for predicting ICU admission and critical care outcomes in ED patients.	Focused on general critical care, not hypertension specific
Chowdhury et al. (2022)	Random Forest, Logistic Regression	Systematic review of hypertension prediction studies	Random Forest and Logistic Regression showed strong performance in predicting hypertension, with age and blood pressure as key predictors, though studies are mostly limited to general populations with few ED-specific datasets.	General population studies; limited ED-specific data
Purushotham et al. (2018)	Hyperparameter-tuned ML	MIMIC-III	Neural networks captured complex interactions but required large data.	High computational complexity; retrospective data only
This Study	Hyperparameter-tuned ML model	Kaggle + RIPAS	Aims to generalize across heterogeneous datasets for hypertension prediction.	Addresses the hypertension-specific gap.

2.6 Critical Remarks of Previous Works

Overall, the existing literature demonstrates that machine learning can predict hypertension, but mostly in outpatient and general population settings where data quality and volume are relatively

favorable. Ensemble methods perform well (Chowdhury et al., 2022), but their application to ED triage data remains underdeveloped. Deep learning approaches have shown promise on large datasets such as MIMIC-III (Purushotham et al., 2018), but the data requirements are substantial, well beyond what is available in a single-center ED context. Emergency department ML work has tended to prioritize ICU admission and critical care outcomes (Raita et al., 2019) over condition-specific prediction. The present study was designed to fill that space: ensemble methods, applied to local ED data, with SHAP-based interpretability built in from the start.

Chapter 3: Methodology

This chapter describes the research design, data sources, and the full processing pipeline developed for predicting hypertension risk from Emergency Department triage data.

3.1 Research Framework and Design

The study was structured across two phases:

Phase 1: Model Benchmarking and Selection: The aim of the first phase was to identify which machine learning algorithm performed most reliably for hypertension prediction before committing to a specific approach for the local dataset. Multiple standard classifiers were trained and evaluated on combined public datasets sourced from Kaggle, which varied considerably in size and structure. RandomizedSearchCV was used to tune hyperparameters for each model, enabling a fair comparison without relying on manual parameter selection. The best-performing model from this phase was taken forward into Phase 2.

Phase 2: Optimization and Local Validation: This phase constitutes the core contribution of the research. The selected models were subjected to deeper hyperparameter tuning and threshold optimization, then evaluated on the RIPAS dataset. The RIPAS data presented specific challenges not encountered in the Kaggle datasets: severe class imbalance (approximately 96% non-hypertensive) and a relatively small total record count of 1,636. These constraints shaped many of the methodological decisions made in this phase, including the choice of evaluation metrics and the decision to apply SMOTE only to training data.

3.2 Data Sources and Description

Kaggle Datasets: Four publicly available datasets relating to cardiovascular health and hypertension were combined, producing a dataset exceeding 10,000 instances. Variables included demographic data such as age, blood pressure, cholesterol, and BMI, along with a binary hypertension label. These datasets were used exclusively during Phase 1 for model selection purposes.

RIPAS Dataset: This dataset comprises 1,636 de-identified ED patient records from RIPAS Hospital. It includes demographic information, vital signs, chief complaint, and comorbidity flags. Working with this dataset required particular care given its size and the pronounced imbalance between hypertensive and non-hypertensive cases. The dataset was used for all Phase 2 development and evaluation.

3.3 Requirement Specifications

3.3.1 User Requirements

- The system should predict hypertension risk at the point of ED triage.
- Outputs should be interpretable and meaningful to clinicians without machine learning expertise.
- The system should function using structured triage data already collected during standard ED workflows.

3.3.2 System Requirements

A minimum of 8 GB RAM (16 GB recommended for efficient cross-validation) is required, with a multi-core CPU such as an Intel i5 or equivalent. GPU support is not required, as XGBoost and

Gradient Boosting are CPU-efficient. Access to a cloud or high-performance computing environment is optional but beneficial for faster training runs.

Python 3.8 or above is required, with development conducted in Jupyter Notebook or a comparable IDE. Required libraries include pandas and numpy for data handling, scikit-learn for modelling and preprocessing, imbalanced-learn for SMOTE, xgboost for model implementation, shap for interpretability, and matplotlib and seaborn for visualization.

3.4 Pre-model Development

3.4.1 Data Preprocessing

Data cleaning was carried out prior to modelling. Variables with more than 30% missing values were excluded from the analysis. For the remaining variables, numerical missingness was addressed using median imputation and categorical missingness using mode imputation, both applied via scikit-learn's SimpleImputer. Values that were clinically implausible, such as systolic blood pressure readings of zero, were treated as missing and imputed accordingly.

Categorical variables including gender, residence type, and smoking status were converted to numerical form using one-hot encoding. All numerical features were subsequently standardized using StandardScaler, which was particularly important for models sensitive to feature magnitude, such as Logistic Regression and SVM.

3.4.2 Data Splitting and Imbalance Handling

The preprocessed dataset was divided into a 70% training set and a 30% test set, with stratification on the target variable to preserve the original class distribution across both subsets. To address the class imbalance, which was particularly severe in the RIPAS dataset, SMOTE was applied to the

training data only, generating synthetic instances for the hypertensive minority class. The test set was kept entirely untouched to ensure it remained representative of real-world conditions.

3.5 Descriptive Analysis

The descriptive analysis of the RIPAS dataset revealed a broad adult age range (18–89 years) with a slight male majority (52%). Mean systolic blood pressure of 138.5 mmHg and diastolic of 84.7 mmHg indicate that a substantial proportion of patients were already in pre-hypertensive or hypertensive ranges at the point of triage, reinforcing the clinical case for systematic screening during this stage of care.

Table 3.5.1 Descriptive Analysis of RIPAS dataset

Variable	Mean / Count	Std / %	Min	Max
Age (years)	52.4	16.8	18	89
Sex (Male)	1,230	52%	–	–
Number of ED visits	3.2	1.9	0	12
Number of admissions	0.8	1.2	0	6
Heart rate (bpm)	82.1	15.7	55	142
Systolic BP (mmHg)	138.5	21.3	90	210
Diastolic BP (mmHg)	84.7	13.5	50	130
Respiratory rate (bpm)	18.9	3.2	12	28
Oxygen saturation (%)	96.2	2.8	85	100
Temperature (°C)	36.8	0.9	35.2	39.5

3.6 Model Development on RIPAS Dataset

In Phase 2, XGBoost and Gradient Boosting were developed and evaluated on the RIPAS dataset using a complete machine learning pipeline designed to ensure reproducibility and prevent data leakage. The pipeline included median and mode imputation, one-hot encoding, feature scaling, and SMOTE applied with a sampling strategy of 0.8, all restricted to training data. Three model variants were evaluated: XGBoost Classifier, Gradient Boosting Classifier, and a hybrid ensemble combining both.

3.6.1 Hyperparameter Tuning

Two complementary tuning strategies were used. RandomizedSearchCV was applied first for broad exploration of the parameter space, followed by GridSearchCV for finer tuning around the most promising regions. A 10-fold stratified cross-validation strategy was used throughout to ensure robust and balanced evaluation across class distributions. Recall was set as the primary optimization metric, reflecting the clinical priority of minimizing false negatives, cases where a hypertensive patient is incorrectly classified as non-hypertensive.

3.6.2 Threshold Tuning

The default classification threshold of 0.5 was lowered to 0.35 following evaluation of the precision-recall trade-off. This adjustment was made to improve recall at the cost of some precision, which is an accepted trade-off in clinical screening contexts where a missed diagnosis carries greater risk than a false positive.

Chapter 4: Experiments, Results and Discussion

4.1 Evaluation criteria

The model evaluation was conducted using five-fold cross-validation to ensure generalization and robustness of performance across unseen data. The following key metrics were used:

- Accuracy (CV Accuracy): Measures the proportion of correctly classified instances.
- Best CV Score: Represents the highest cross-validation performance obtained during hyperparameter tuning.
- ROC-AUC (Receiver Operating Characteristic – Area Under Curve): Assesses the model's ability to distinguish between classes, with 1.0 indicating perfect discrimination.

Accuracy and ROC-AUC were both used to evaluate model performance. In highly imbalanced clinical datasets, which are common in medical screening and diagnostic applications, accuracy can be misleading as it may be disproportionately influenced by the majority class. In contrast, ROC-AUC provides a threshold-independent measure of a model's ability to discriminate between classes and is widely recommended for assessing binary clinical prediction models (Fawcett, 2006; Hanley & McNeil, 1982). Furthermore, research in machine learning evaluation highlights that ROC-AUC remains a robust metric for assessing discriminatory performance in diagnostic systems, particularly when class imbalance is present (Saito & Rehmsmeier, 2015; Mandrekar, 2010). Therefore, ROC-AUC was prioritized alongside accuracy to provide a more reliable and clinically meaningful evaluation of model performance.

Given the class imbalance, recall and ROC-AUC were prioritized over accuracy, as they better reflect model performance in clinical screening contexts where minimizing false negatives is critical.

4.2 Experimental Model Performance on Kaggle Datasets (Preliminary Results)

Initial experiments were conducted on four aggregated Kaggle datasets. It is important to note the nature of the target variable across these datasets:

- Datasets 2 and 3 contained a directly measured, clinically defined hypertension column.
- Datasets 1 and 4 did not have a direct hypertension label. For these, a hypertension target variable was engineered using the JNC-7 guidelines-based equation:

  Hypertension = (Systolic_BP >= 140) | (Diastolic_BP >= 90).

The baseline models, including Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting, Multi-Layer Perceptron (MLP), and XGBoost, were optimized using RandomizedSearchCV and evaluated.

***Table 4.2.1**:* Model Performance on Kaggle Dataset 2

Model	**CV Accuracy**	**search.best_score()**	**CV ROC-AUC**
XGBoost	**0.824**	0.877	**0.893**
LogisticRegression	0.817	**0.88**	**0.893**
GradientBoosting	0.821	0.879	0.892
SVM	0.802	0.871	0.884
RandomForest	0.791	0.867	0.881
MLP	0.789	0.858	0.857
DecisionTree	0.771	0.827	0.847

***Table 4.2.2**:* RandomizedSearchCV Best Parameters used on Kaggle Dataset 2

Model	**Parameter**	**Best Value**
XGBoost	colsample_bytree	0.911762697
	learning_rate	0.125512955
	max_depth	4

	min_child_weight	1
	n_estimators	61
	subsample	0.857463052
LogisticRegression	C	0.07825733
	solver	liblinear
GradientBoosting	learning_rate	0.076335324
	n_estimators	97
SVM	C	0.414463876
	kernel	rbf
RandomForest	max_depth	5
	min_samples_leaf	2
	min_samples_split	7
	n_estimators	114
MLP	alpha	0.006822786
	hidden_layer_sizes	(100,)
DecisionTree	max_depth	4
	min_samples_leaf	4
	min_samples_split	8

Table 4.2.3: Model Performance on Kaggle Dataset 3

Model	**CV Accuracy**	**search.best_score()**	**CV ROC-AUC**
SVM	0.677	**0.792**	**0.779**
LogisticRegression	0.709	0.8	0.777
XGBoost	0.82	0.796	0.754
RandomForest	0.773	0.793	0.753
GradientBoosting	0.847	0.788	0.731
MLP	**0.887**	0.733	0.718
DecisionTree	0.636	0.724	0.717

Table 4.2.4: RandomizedSearchCV Best Parameters used on Kaggle Dataset 3

Model	Parameter	Best Value
SVM	C	1.934347899
	kernel	linear
LogisticRegression	C	0.57411579
	solver	saga
XGBoost	colsample_bytree	0.895266542
	learning_rate	0.061282316
	max_depth	4
	min_child_weight	2
	n_estimators	51
	subsample	0.75454749
RandomForest	max_depth	5
	min_samples_leaf	3
	min_samples_split	7
	n_estimators	104
GradientBoosting	learning_rate	0.046686958
	n_estimators	121
MLP	alpha	0.004692489
	hidden_layer_sizes	(50,)
DecisionTree	max_depth	4
	min_samples_leaf	3
	min_samples_split	9

Performance on Datasets 2 and 3, which represent a more realistic and complex clinical definition of hypertension, was strong but more moderate. This suggests the models are effectively learning genuine, non-trivial patterns from a broader set of features.

***Table 4.2.5**: Model Performance on Kaggle Dataset 1*

Model	CV Accuracy	search.best_score()	CV ROC-AUC
DecisionTree	**1.000000**	**1.000000**	**1.000000**
XGBoost	**1.000000**	**1.000000**	**1.000000**
GradientBoosting	**1.000000**	**1.000000**	**1.000000**
RandomForest	0.898185	0.996111	0.992896
LogisticRegression	0.955012	0.990631	0.990775
SVM	0.947119	0.989578	0.989199
MLP	0.937648	0.983387	0.983559

***Table 4.2.6**: RandomizedSearchCV Best Parameters used on Kaggle Dataset 1*

Model	Parameter	Best Value
LogisticRegression	C	0.590836122
	solver	saga
DecisionTree	max_depth	5
	min_samples_leaf	2
	min_samples_split	7
RandomForest	max_depth	5
	min_samples_leaf	2
	min_samples_split	7
	n_estimators	121
SVM	C	0.75051593
	kernel	linear
GradientBoosting	learning_rate	0.084908024
	n_estimators	142
MLP	alpha	0.00975632
	hidden_layer_sizes	(100,)
XGBoost	colsample_bytree	0.812362036
	learning_rate	0.240142861
	max_depth	5
	min_child_weight	4

	n_estimators	110
	subsample	0.879055047

Table 4.2.7: *Model Performance on Kaggle Dataset 4*

Model	CV Accuracy	search.best_score()	CV ROC-AUC
GradientBoosting	**0.988**	**1**	**1**
XGBoost	0.986	0.998	0.999
DecisionTree	0.973	0.978	0.99
RandomForest	0.91	0.99	0.986
SVM	0.932	0.972	0.982
MLP	0.929	0.974	0.978
LogisticRegression	0.913	0.953	0.962

Table 4.2.8*: RandomizedSearchCV Best Parameters used on Kaggle Dataset 4*

Model	Parameter	Best Value
GradientBoosting	learning_rate	0.046686958
	n_estimators	121
XGBoost	colsample_bytree	0.895266542
	learning_rate	0.061282316
	max_depth	4
	min_child_weight	2
	n_estimators	51
	subsample	0.75454749
DecisionTree	max_depth	5
	min_samples_leaf	4
	min_samples_split	7
RandomForest	max_depth	5
	min_samples_leaf	2
	min_samples_split	7
	n_estimators	121

SVM	C	7.319987723
	kernel	rbf
MLP	alpha	0.002223391
	hidden_layer_sizes	(50,)
LogisticRegression	C	0.57411579
	solver	saga

The models performed exceptionally well on Datasets 1 and 4, with several achieving perfect CV scores. This is likely because the target variable was directly derived from the BP features used as inputs, making the prediction task less complex, resulting in artificially inflated performance due to label leakage.

Interpretation of Preliminary Results:

This distinction highlights the importance of using clinically defined targets when evaluating model performance, as engineered targets may artificially inflate results and not reflect real-world predictive capability.

4.3 Final Model Performance on RIPAS dataset

4.3.1 Overview

The performance of XGBoost, Gradient Boosting, and a hybrid ensemble model was evaluated under three conditions:

- Baseline (no tuning)
- RandomizedSearchCV / GridSearchCV
- Threshold adjustment (0.35)

4.3.2 Key Findings

4.3.2.1 XGBoost Performance

XGBoost consistently demonstrated strong and stable performance across all evaluation steps. Threshold tuning significantly improved recall, allowing better identification of hypertensive patients, although with the slight drop in overall accuracy. Its robustness and balance across metrics justified its selection as the final model.

4.3.2.2 Gradient Boosting Performance

Gradient Boosting showed moderate baseline performance but responded strongly to threshold adjustments, with significant improvements in recall. However, it exhibited less stability compared to XGBoost, particularly in maintaining overall performance consistency.

4.3.2.3 Hybrid Model Performance

The hybrid ensemble model maintained competitive performance but did not consistently outperform XGBoost. While recall improved after tuning, the added complexity of ensembling provided limited practical benefit for this task.

Table 4.3.1: *Model Performance Comparison on RIPAS dataset*

Model	Stage	Threshold	Accuracy	ROC-AUC	Recall	Precision	F1-score
XGBoost	Baseline	0.5	0.96	0.96	0.67	0.46	0.55
	RandomCV	**0.35**	0.95	0.945	**0.89**	0.42	0.57
	GridSearch	0.5	0.95	0.96	0.67	0.40	0.50
Gradient Boosting	Baseline	0.50	0.94	0.95	0.44	0.27	0.33
	RandomCV	**0.35**	0.89	0.92	**0.89**	0.24	0.37
	GridSearch	0.50	0.90	0.92	0.78	0.25	0.38
Hybrid	Baseline	0.50	0.96	0.96	0.67	0.46	0.55
	Tuned	0.50	0.94	0.94	0.78	0.33	0.47

4.3.3 Interpretation of Results

The tuned XGBoost model demonstrated improved performance compared to the baseline model, particularly in terms of recall, which is critical in clinical applications. A similar trend was observed for Gradient Boosting, which also showed competitive performance after optimization.

- Recall improvement indicates the model's enhanced ability to correctly identify patients with hypertension, reducing the risk of missed diagnoses. Both XGBoost and Gradient Boosting achieved strong recall performance, reinforcing their suitability for clinical screening tasks.
- ROC-AUC improvement reflects better overall discrimination between hypertensive and non-hypertensive patients. XGBoost slightly outperformed Gradient Boosting, although both models demonstrated high discriminative ability.

- The relatively stable accuracy across both models suggests that performance gains were achieved without compromising overall prediction correctness.

Overall, while Gradient Boosting performed competitively, XGBoost provided more consistent and slightly superior results across multiple evaluation metrics, leading to its selection as the final model for deployment.

These findings confirm that hyperparameter tuning and class imbalance handling (SMOTE) significantly enhanced model performance in a real-world clinical dataset.

4.4 Feature Importance and Model Interpretability

4.4.1 Feature Importance and Model Interpretability in Kaggle Datasets

Complementary SHAP (Shapley Additive Explanations) analysis was conducted on the best-performing model (e.g., XGBoost) to ensure transparency and identify the most influential predictors. This reinforces the clinical validity of the model, as it relies on physiologically meaningful features rather than spurious correlations.

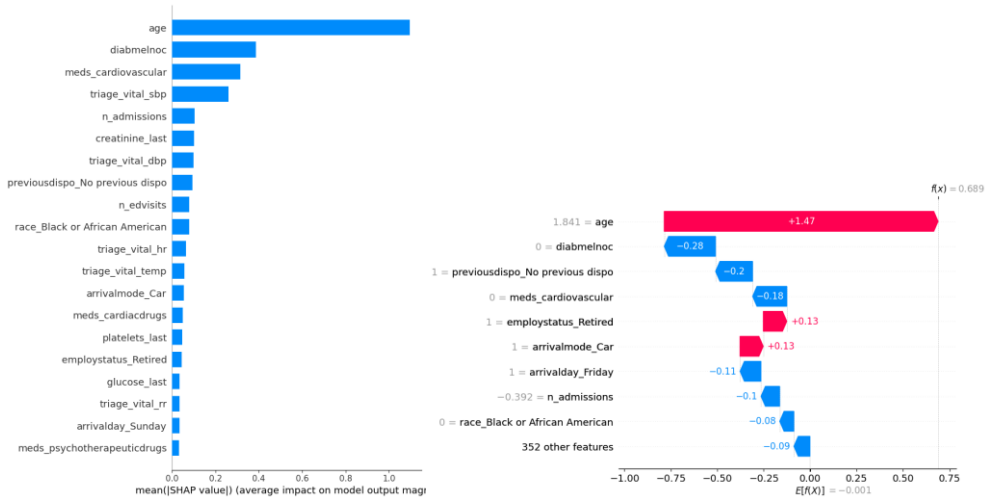

Figure 4.4.1 *SHAP analysis on Dataset 2*

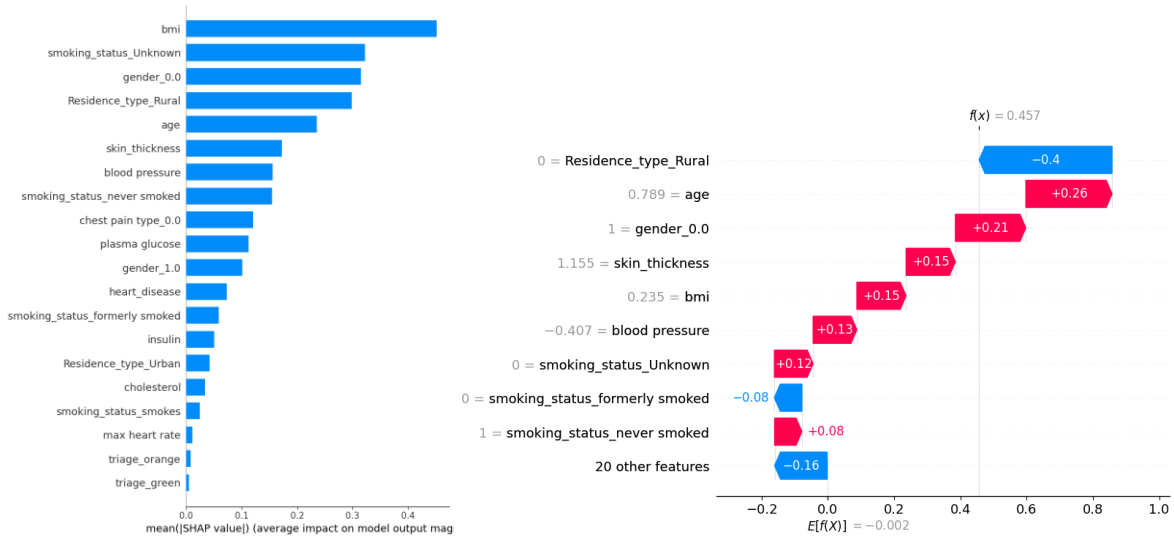

Figure 4.4.2 *SHAP analysis on Dataset 3*

For datasets with the clinical target (2 & 3), the analysis revealed a more diverse set of top features, including Age, Cholesterol_level, and BMI, alongside blood pressure metrics. This aligns with clinical understanding and validates the model's ability to identify multifaceted risk factors.

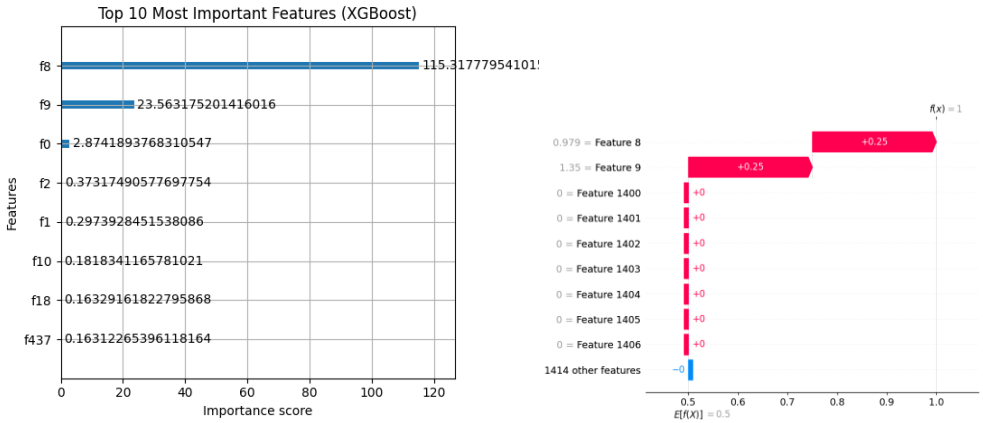

Figure 4.4.3 *SHAP analysis on Dataset 1*

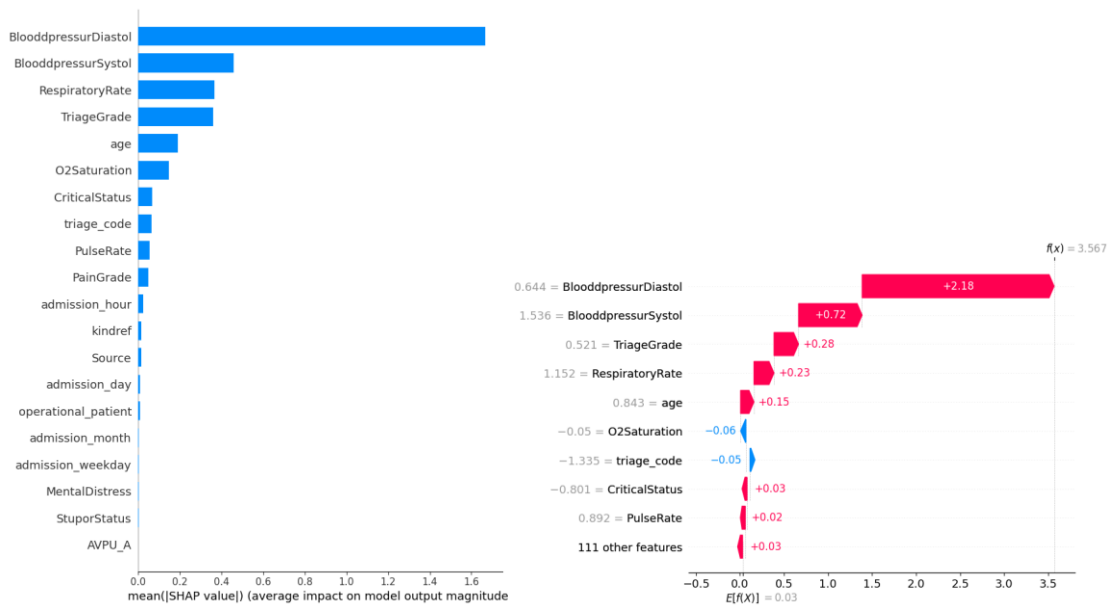

Figure 4.4.4 *SHAP analysis on Dataset 4*

For datasets with the engineered target (1 & 4), Systolic_BP and Diastolic_BP were, as expected, the dominant features.

4.4.2 Feature Importance and Model Interpretability on RIPAS Dataset

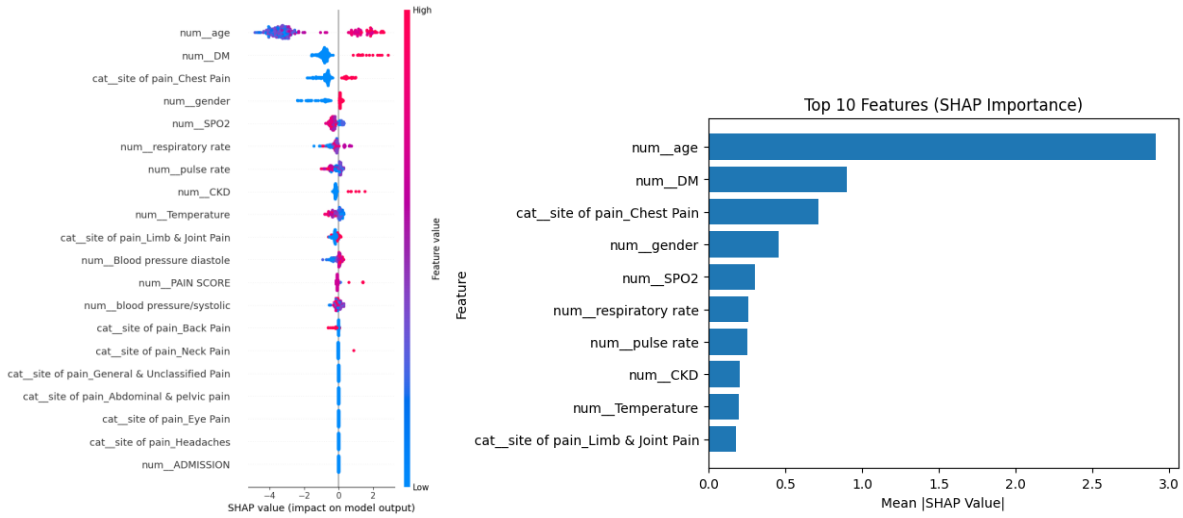

Figure 4.4.5 *SHAP analysis on RIPAS dataset*

SHAP analysis of the RIPAS dataset shown in Figure 4.4.5 indicates that age is the strongest predictor of hypertension risk. Comorbidities such as diabetes mellitus (DM) and chronic kidney disease (CKD) also contribute significantly. Clinical features such as chest pain and vital signs (SpO_2, respiratory rate, pulse rate) have a moderate influence. Additionally, blood pressure variables are less dominant, indicating the model captures a broader risk profile. Overall, the results align with clinical knowledge and support model interpretability.

4.5 Discussion and Key Findings

The experiments across both phases produced a fairly clear picture of where machine learning works well for hypertension prediction, where it struggles, and what that means for the RIPAS context specifically.

4.5.1 Performance on Clinically-Defined vs. Engineered Targets

The most striking pattern across the Kaggle experiments was how dramatically performance varied depending on how the hypertension label was defined. On Datasets 1 and 4, where the target

variable was derived directly from blood pressure thresholds, essentially flagging anyone with systolic BP ≥ 140 or diastolic BP ≥ 90, several models achieved near-perfect scores. At first glance this looks impressive, but it is actually a sign that something has gone wrong methodologically. The models were not learning to predict hypertension from a range of features; they were learning to re-derive the same rule that was already encoded in the input data. This is label leakage, and the inflated scores it produces are not meaningful in a real predictive sense.

The results from Datasets 2 and 3 were more informative precisely because they were more modest. These datasets used a clinically recorded hypertension diagnosis rather than an engineered threshold, which means the prediction task was genuinely harder and the results were more trustworthy. Performance was still strong, ROC-AUC values in the range of 0.75 to 0.89, but not suspiciously so. This contrast reinforced the decision to use the RIPAS dataset as the primary validation context, since it carries real clinical diagnoses from actual patient records.

4.5.2 Model Selection and Robustness

Across all four Kaggle datasets, ensemble methods consistently came out ahead of simpler approaches, with XGBoost and Gradient Boosting delivering the most reliable results. Their ability to capture non-linear relationships between features and handle mixed data types made them a natural fit for the kind of heterogeneous clinical data found in triage records. One nuance worth noting is that on Dataset 2, Logistic Regression matched XGBoost almost exactly on ROC-AUC (0.893 vs 0.893), which suggests that for cleaner, well-structured clinical datasets, simpler models are still competitive. Even so, ensemble methods were taken forward to Phase 2 because of their overall consistency, they did not rely on any one dataset having particular properties to perform well.

4.5.3 Interpretability and Clinical Relevance

SHAP analysis added something important beyond the raw performance numbers. On the clinically-defined Kaggle datasets, the features that drove predictions the most, age, cholesterol, blood pressure, are all well-established hypertension risk factors, which gives some confidence that the model is picking up on real clinical patterns rather than artefacts in the data. On the RIPAS dataset, the picture was slightly different and arguably more interesting. Age was the strongest predictor by some distance, followed by diabetes mellitus and chronic kidney disease. Raw blood pressure readings, which you might expect to dominate, were actually less influential than the comorbidity and demographic features, suggesting the model was building a broader risk profile rather than essentially re-applying the JNC-7 threshold rule. Vital signs like SpO_2, respiratory rate, and pulse rate also contributed, though at a more moderate level.

This matters not just as a technical validation exercise but because any model that gets deployed in a clinical setting needs to be explainable. A triage nurse or clinician is not going to act on a black-box risk score, they need to see which patient characteristics are driving the flag before they can engage with it meaningfully.

4.5.4 Ripas Model Performance

On the RIPAS dataset, the tuned XGBoost model with threshold adjustment to 0.35 achieved a recall of 0.89 and ROC-AUC of 0.945. These are strong numbers given the dataset constraints, 1,636 records, approximately 96% non-hypertensive, with all the imputation and SMOTE choices that entailed. The recall improvement from baseline (0.67) to the tuned model (0.89) is the most clinically meaningful result in the study, because in a screening context the cost of missing a hypertensive patient is considerably higher than the cost of flagging someone for a follow-up check that turns out to be unnecessary.

The precision figure of 0.42 is admittedly low, and it is worth being direct about what that means in practice: for every hypertensive patient correctly identified, roughly one non-hypertensive patient would also be flagged. Whether that trade-off is acceptable depends on the clinical workflow it would be embedded in, how much capacity exists to follow up on flagged cases, and what the downstream cost of a false positive actually is in that setting. Managing that balance carefully would need to be a central consideration in any future deployment work.

4.6 Answering the Research Questions

- RQ1: "Can experimental, baseline ML models effectively predict hypertension from triage data?" Based on the high ROC-AUC scores achieved by the top models on clinical data, the results strongly suggest that they can. The performance on datasets with a clinical definition of hypertension confirms the feasibility of this approach for Phase 2 development on the RIPAS dataset.
- RQ2: “Which features are most important in driving the prediction?” The answer differs by dataset. On the Kaggle clinical datasets (Datasets 2 and 3), SHAP analysis identified Age, Systolic BP, Diastolic BP, and Cholesterol as the top predictive drivers. On the RIPAS dataset, Age, comorbidities (diabetes mellitus and chronic kidney disease), and clinical vital signs (SpO_2, respiratory rate, pulse rate) were most predictive, with blood pressure variables being less dominant. This confirms that the model captures a broader, clinically meaningful risk profile beyond blood pressure alone.

4.7 Limitations of the Study

While this study demonstrates strong performance, several limitations remain. First, the use of engineered target variables in some Kaggle datasets may have introduced label leakage, resulting in overly optimistic performance. Second, although the models were validated on the RIPAS

dataset, the data is sourced from a single hospital, which may limit generalizability. Third, the study relies on retrospective data, and real-time deployment may present additional challenges. Finally, only structured triage data was used, excluding potentially valuable unstructured clinical information.

4.8 Ethical Considerations

The study used anonymized data. The potential for model bias was mitigated by using stratification. The use of SHAP ensures transparency for future clinical evaluation. When deploying a model that uses derived labels, it is crucial to ensure it does not reinforce simplistic diagnostic criteria and is validated against true clinical diagnoses.

4.9 Next Steps and Future Work

- External validation using multi-hospital datasets
- Integration into real-time ED triage systems
- Exploration of temporal models (e.g., LSTM) for longitudinal patient data
- Inclusion of unstructured clinical notes using NLP.

Chapter 5: Conclusion

This study investigated whether machine learning could predict hypertension risk from Emergency Department triage data at RIPAS Hospital, where hypertension was the most common comorbidity yet was not systematically screened for at triage. The goal was a practically useful model that could integrate into existing clinical workflows.

XGBoost delivered the strongest performance, achieving a recall of 0.89 and ROC-AUC of 0.945 on the RIPAS dataset after hyperparameter tuning and threshold optimization, meaningfully reducing missed hypertension cases at triage.

The RIPAS dataset presented real-world challenges: severe class imbalance and limited sample size. These required careful methodological choices, including SMOTE applied only to training data, stratified cross-validation, and recall-focused tuning. Preliminary benchmarking on Kaggle datasets also surfaced the risk of label leakage, reinforcing the importance of clinically grounded outcome definitions.

SHAP-based interpretability analysis confirmed that predictions were driven by clinically meaningful features such as age, diabetes mellitus, chronic kidney disease, and vital sign patterns rather than spurious correlations, which is considered essential for any future clinical deployment.

Key limitations include reliance on retrospective single-center data and the exclusion of unstructured clinical information. Future work should focus on external validation, real-time triage integration, and longitudinal modelling using temporal patient data.

Overall, these results demonstrate that machine learning, carefully developed and validated against real clinical constraints, has genuine potential to support earlier hypertension detection and improve ED triage decision-making.

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